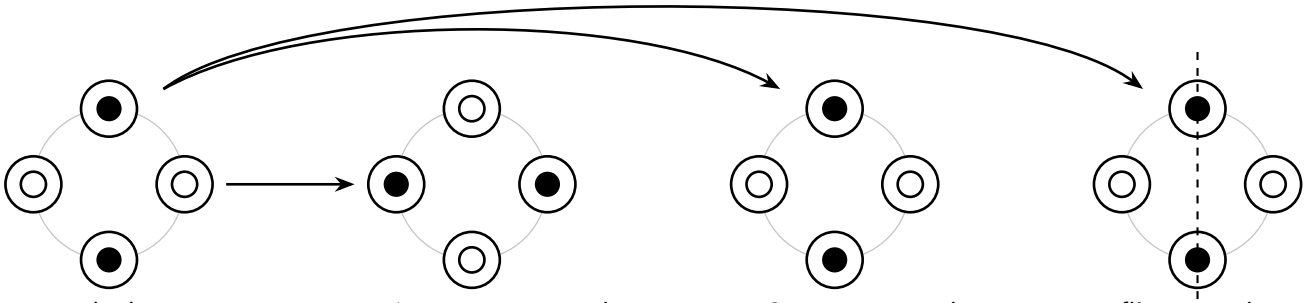
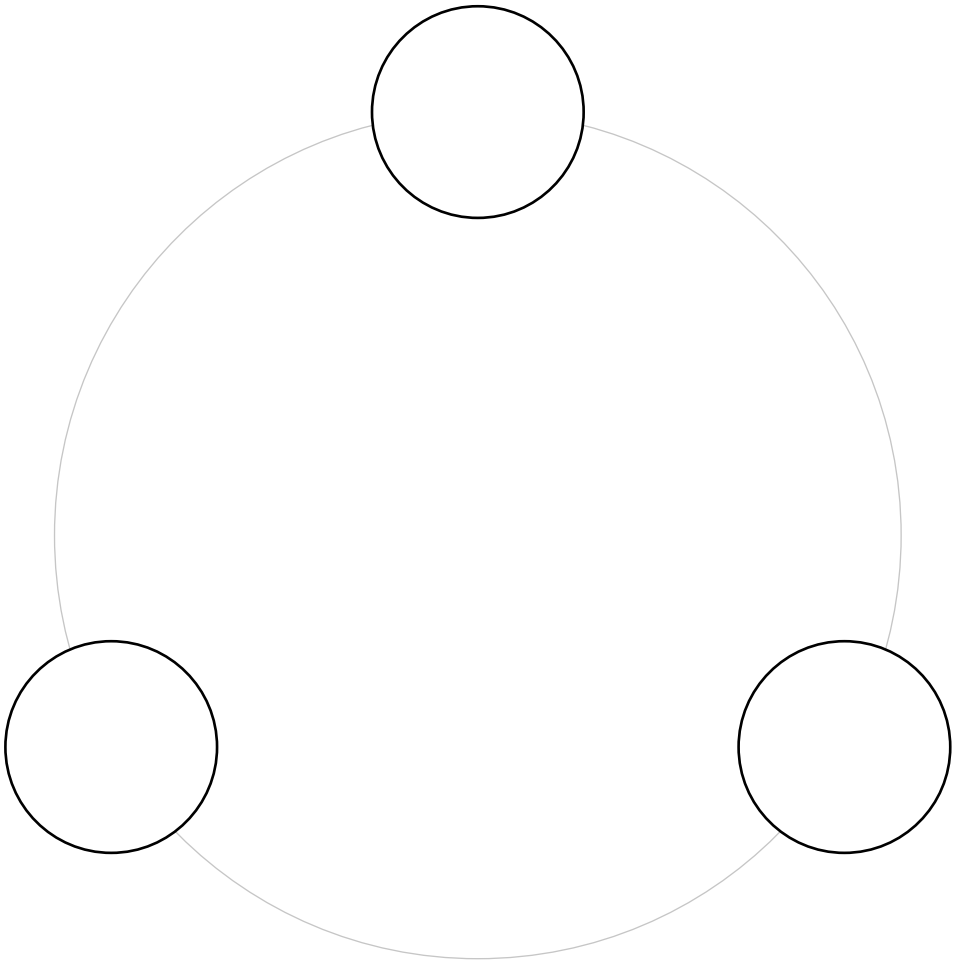
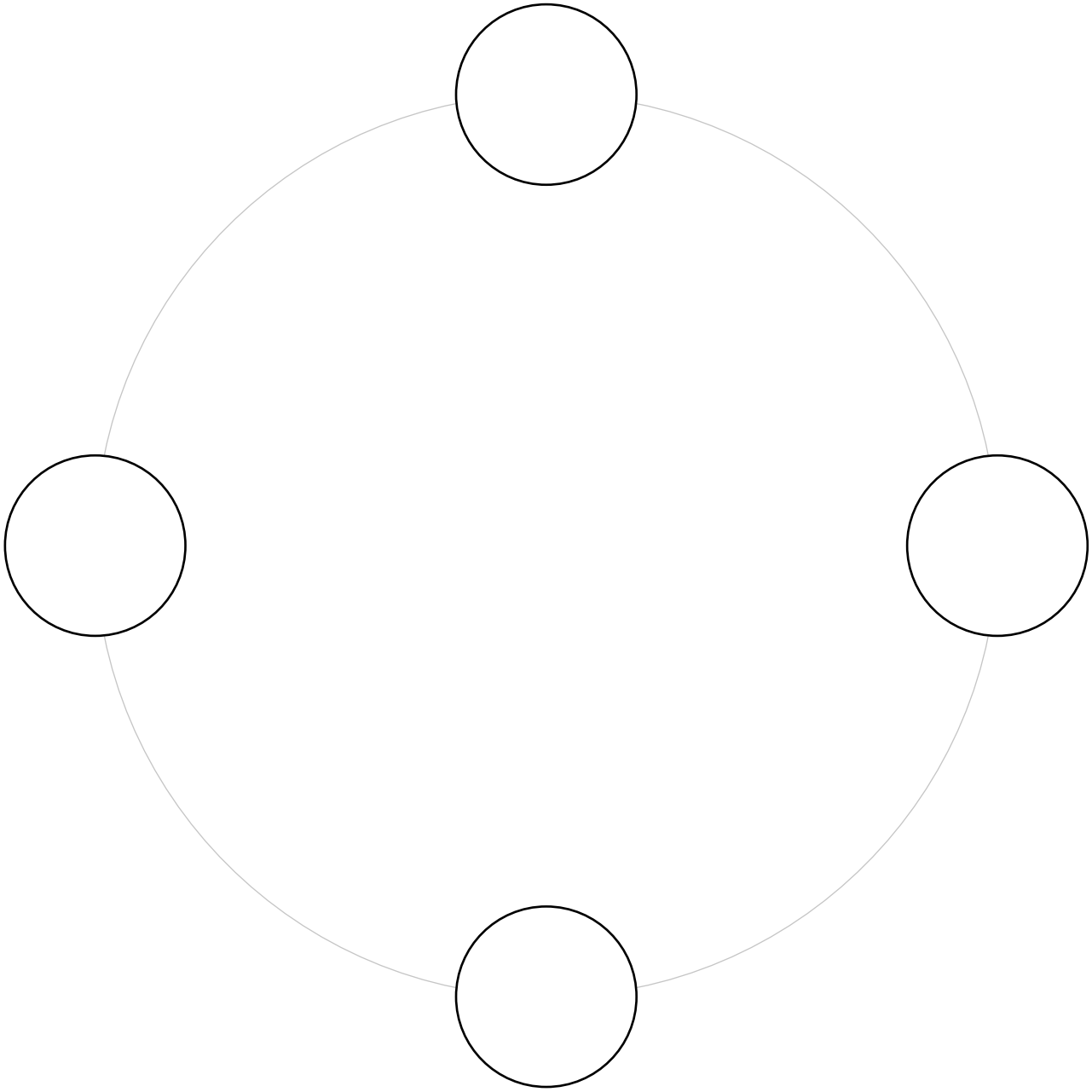
Fill every spot; neighbors may be the same kind. Copy a pattern on tracing paper with a dot of each counter's color. One partner builds. The other keeps the original still and turns or flips (turns over) the copy to a new place. If every dot sits on its own color, that is a match. Putting the copy back where it started, or turning it all the way around, does not count. A flip may be over any line through the middle.



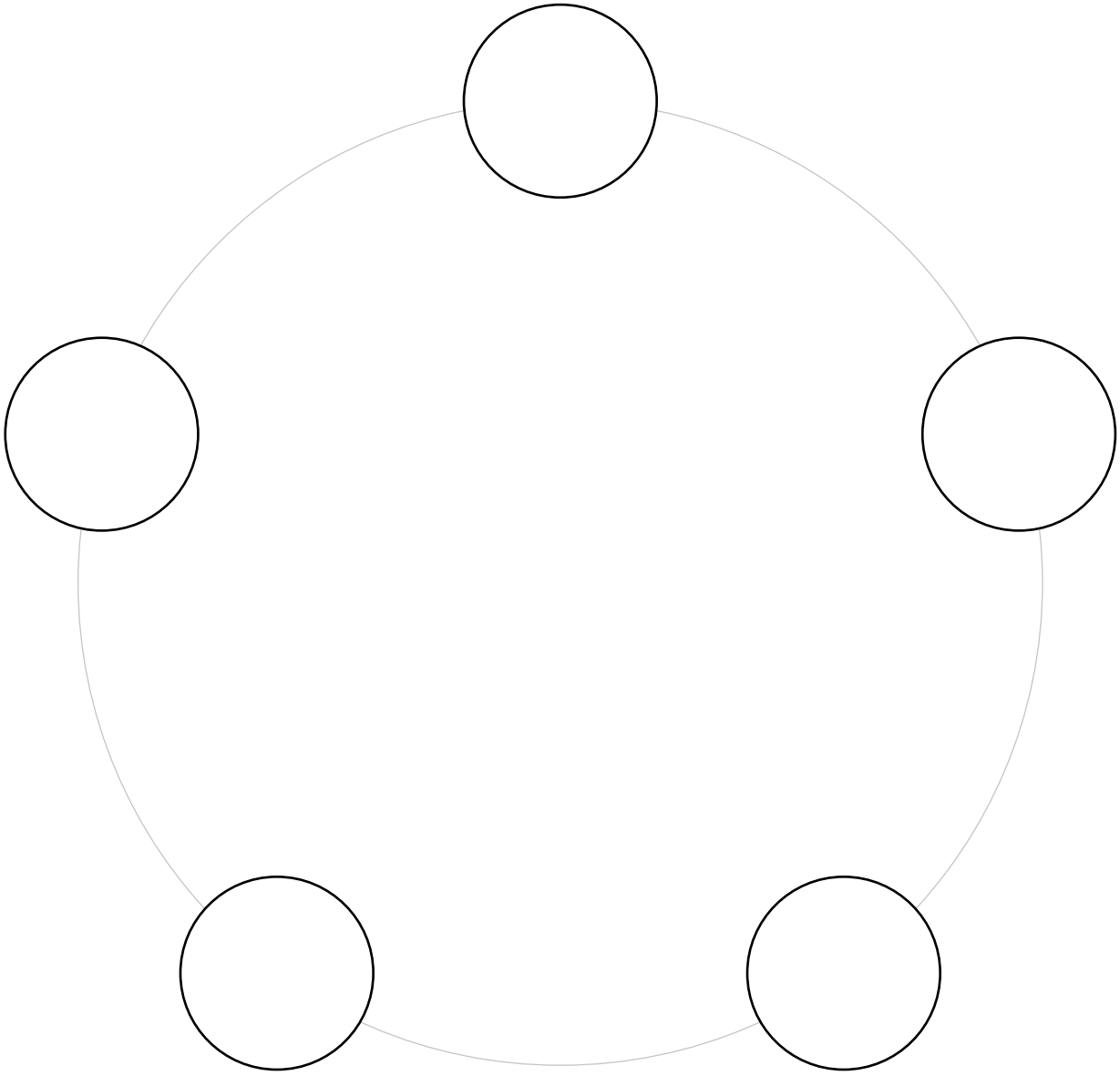
Problem 1: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match.



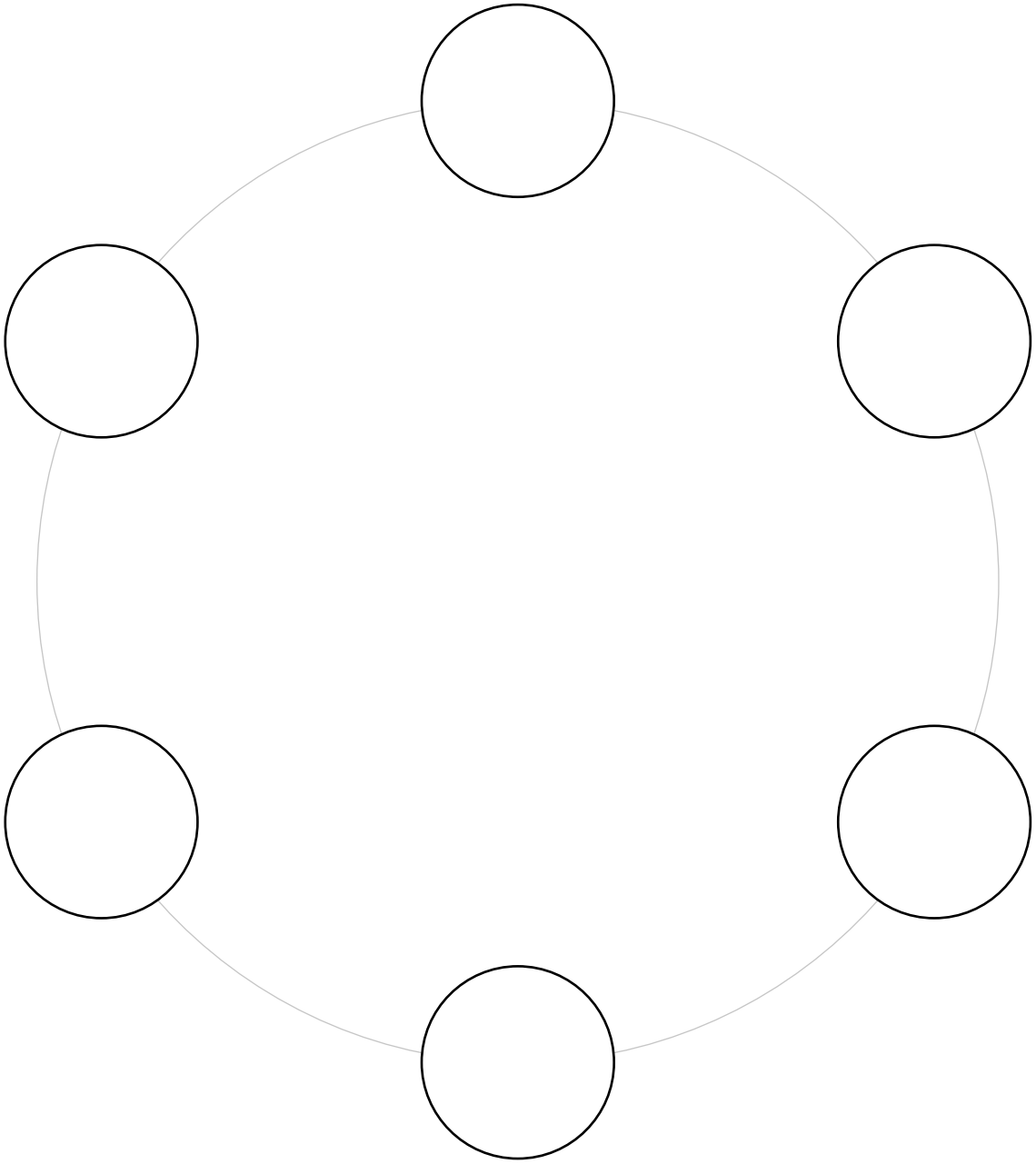
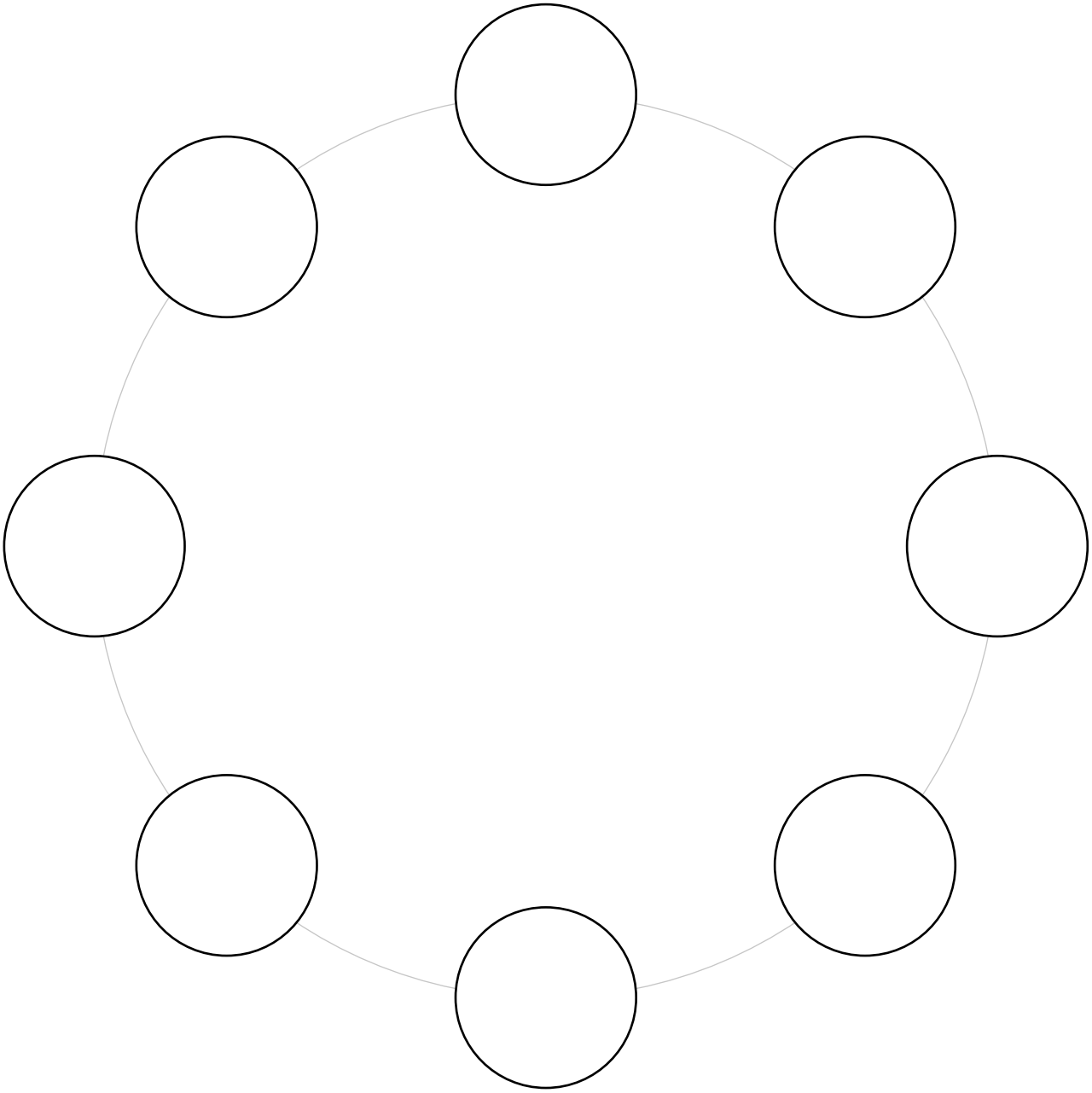
Problem 2: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match.



Problem 3: Find how few kinds of counter can make a pattern on this ring that no turn or flip can match. Can you explain why fewer kinds would fail?



Problem 4: Find a six-ring pattern using only two kinds of counter that no turn or flip can match. Explain how you know no turn or flip can match it.



Problem 5: Find an eight-ring pattern with two kinds of counter that no turn or flip can match. How few counters of one kind can it have?